

Filter Overview

So What is a Filter

A filter is some kind of circuit (analog) or linear time-invariant system (digital) whose transfer function we are interested in applying to a signal. Typically, this is for the rejection or enhancement of some frequency/range of frequencies. There are an assortment of filter responses and types that we can use.

Frequency Response

Filters are typically used to remove a range of frequencies from a signal, and are (usually) named for the frequencies that they allow to pass. The most commonly seen types:

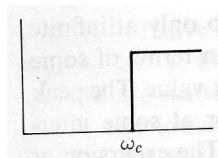
Lowpass

A lowpass filter removes high frequencies from a signal. The (amplitude) frequency response of an ideal lowpass filter looks like this:



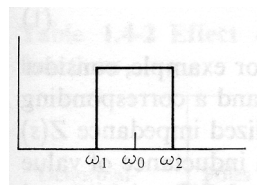
Highpass

A highpass filter does the opposite of a lowpass, removing low frequency content from a signal (surprise!). The frequency response is as such:



Bandpass

A bandpass filter passes a range of frequencies, and removes frequencies outside this range; essentially, it is a highpass filter cascaded with a lowpass filter. The frequency response looks like this:



Bandstop

Alright, guess.

Notch

A notch filter is a bandstop filter with an extremely narrow stopband, typically used when there is a particular frequency causing noise issues (e.g., a notch filter at 60 Hz to remove power line interference).

Cutoff Frequency

This is the frequency describing the operational characteristics of the filter. So a lowpass filter with a cutoff frequency of 100 Hz passes frequencies below 100 Hz and removes frequencies greater than 100 Hz. Bandpass/stop filters specify two cutoff frequencies. The cutoff frequency is also known as the *corner frequency* or *-3-dB frequency* for reasons about to become apparent.

The Ideal Filter Response Does Not Exist

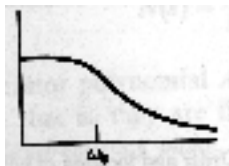
The above frequency responses show an ideal “brick wall” response which does not really exist. The shapes of various types of real filter responses are discussed in the following section. This lack of an ideal filter response is the reason for using the term “-3-dB frequency.” At this frequency, the magnitude of a signal passed through the filter is 3 dB lower than in the passband (recall that -3 dB corresponds to half the original power in the signal). Thus, at the corner frequency, the signal has half the power it originally had. The steepness of the dropoff in the stopband is determined by the *order* of the filter (basically, the more poles/zeros in the frequency response, the higher the filter order). Also, at some point the gain levels off, so there is a gain in the passband, and then a different (obviously, lower) gain in the stopband. The take home message is: **frequencies at and above (or below, etc) the cutoff are attenuated, not removed.** In the stopband of the filter, the gain falls off at a rate of -20 dB/decade (factor of 10) or -6 dB/octave (factor of two) of frequency for each pole in the transfer function of the filter; thus a second-order filter would have a falloff of -40 dB/decade.

Filter Types

Since the ideal filter response does not exist, we must approximate it as best we can. There exist some more popular approximations, described below.

Butterworth

A Butterworth filter characteristic is shown below. The defining characteristic of a Butterworth filter is its so-called *maximally flat* response, i.e., the passband and stopband are completely flat. Thus, all frequencies in the passband are treated exactly the same, which is

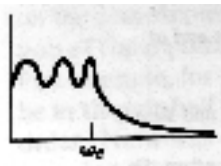


often quite desirable. The problem with a Butterworth filter is that the gain drops off relatively slowly (in the transition regions, that is).

Chebyshev

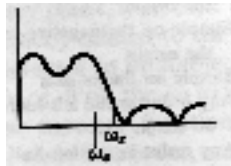
A Chebyshev filter characteristic is shown below. Chebyshev filters (well, Type I Chebyshev filters) are characterized by having an *equiripple* passband, where there is a ripple in the frequency response in the passband, but it is always of the same magnitude (and is one of the design parameters of the filter). While the ripple in the passband of a Chebyshev filter will introduce some distortion into the signal of interest, the frequency response falls off much more steeply for a Chebyshev filter of a given order than for the equivalent Butterworth filter. Note the supposed “equiripple” characteristic is very dependent on a resonance between the components of the filter...thus, the ripple often ain't so equal.

For Type II Chebyshev filters, the above discussion about an equiripple characteristic applies to the stopband.



Elliptic

An elliptic filter response is shown below, and is characterized by having an equiripple passband and stopband, and an even steeper dropoff than the Chebyshev filter.



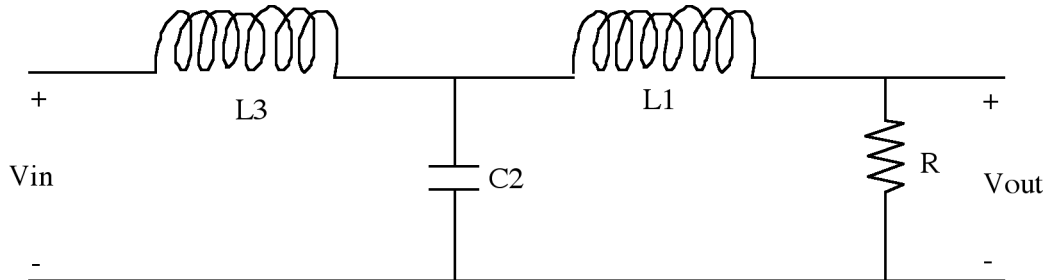
FIR

All of the above are so-called infinite impulse response (IIR) filters, meaning that their response to an impulse input goes on (generally decaying, unless the filter is unstable) to infinity. All these responses can be approximated using DSP, but have some issues with implementation, generally relating to stability and phase issues. Thus, in the digital domain, one can design a finite impulse response (FIR) filter, which avoids some of these issues. An FIR filter, however, will require a higher order to achieve a steep rolloff from the passband than any IIR filter.

Filter Implementation

Passive

Passive filters employ entirely passive components (resistors, capacitors, inductors) to achieve a given filter characteristic. The order of the filter is given by the number of energy-storage circuit elements present.

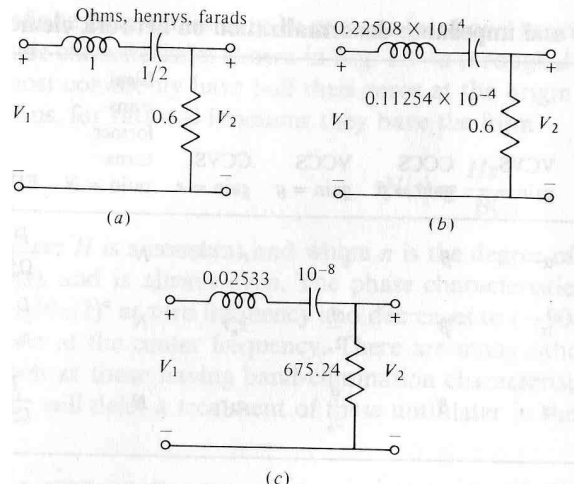


Third order filter. Lowpass or highpass?

Difficulties involved with passive filters involve design involving inductors (issues include mutual inductance, resistance inherent in the inductors, and the fact that one cannot make an integrated inductor on a chip, though such can be approximated). Additionally, since only passive components are used, the gain in the passband will always be less than 1.

Filter Design, Denormalization and Transformations

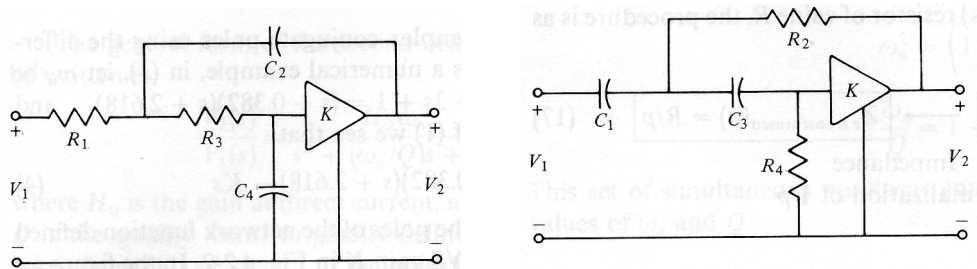
To simplify design calculations, passive filters are typically designed using normalized component values as shown below in (a). Generally speaking, there is a dearth of useful 0.5 F capacitors in the world. Thus, frequency and impedance denormalizations are performed in order to obtain reasonable component values. *Frequency denormalization* changes the corner frequency of the filter, and involves dividing inductor and capacitor values by a constant. *Impedance denormalization* does not change the operation of the filter, but scales all values to (hopefully) something that can realistically be built; it involves multiplying resistor and inductor values by a constant, and dividing capacitor values by the same. In the example shown below, the circuit in (a) is a bandpass filter with a center frequency of $\sqrt{2}$ rad/sec. Applying a frequency denormalization of $(2\pi/\sqrt{2}) \times 10^4$ changes to a center frequency of 10 kHz and gives the values in (b). To obtain a standard capacitor value, say 10 nF, we would then use an impedance denormalization of 1125.4.



Most commonly, designers will design a lowpass filter with a particular set of characteristics, and then transform it to a different type (i.e., highpass, bandpass) filter. This involves transforming the complex frequency variable in the transfer function of the filter. In the lowpass to highpass transformation, the original s (think LaPlace transforms here) becomes p , with the relation $s = 1/p$. It is not difficult to show that an inductor with inductance L becomes a capacitor with capacitance $1/L$, while a capacitor with capacitance C becomes an inductor with inductance $1/C$.

Active

Active filters make use of some active components (typically, op amps) to achieve their goals. The most popular architecture here is the Sallen-Key filter, two examples of which are shown below. Use of active filters removes the need for inductors, as well as providing the capability of having amplification in the passband of the filter.



Sallen Key lowpass and highpass filters

Sallen-Key Filter

The Sallen-Key architecture shown is extremely useful for filter implementation, as the lack of inductors removes any of the inherent difficulties of working with inductors and mutual inductance, while also allowing for construction as an integrated circuit. Implementation of

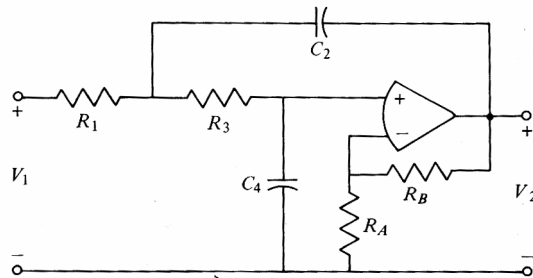
any filter, however, will require some design decisions. For example: for the Sallen-Key lowpass filter using two resistors, two capacitors and an amplifier on the left, above, one can show that the design parameters of this filter are the cutoff frequency ω_c , the DC gain H_0 and the filter Q , and are given by:

$$\omega_n = \frac{1}{\sqrt{R_1 R_3 C_2 C_4}}$$

$$\frac{1}{Q} = \sqrt{\frac{R_3 C_4}{R_1 C_2}} + \sqrt{\frac{R_1 C_4}{R_3 C_2}} + (1 - K) \sqrt{\frac{R_1 C_2}{R_3 C_4}}$$

$$H_0 = K$$

This gives us three equations with five unknowns (four circuit components, and the gain), which leaves us with some design choices to make for a filter; welcome to being boy (or girl) engineer. A common set of constraints to impose is to have equal-valued capacitors and a gain (K) of 2. A common configuration using an op-amp is the following:



This circuit should explain for the decision for using a gain of 2. Based on this, we can now specify a passband gain (in this case, 2), a cutoff frequency and filter Q , and the components of the circuit are determined. For example, for a Butterworth filter using the above parameters, these equations simplify quite nicely.

Filter Application: Touch Tone Phone

